

Electronic Sales Data Analytics (DA) Report

1. Problem Statement

The Electronic Sales business wants to analyze its historical transaction log to evaluate corporate revenue trends, identify gross margin leaks, trace regional fulfillment dynamics, and establish structured, data-driven optimization strategies. The primary goal of this reporting initiative is to enhance profitability across categories and distribution networks by managing reverse-logistics metrics (returns), setting systemic discount caps, and optimizing regional sourcing protocols.

2. Dataset Overview

This dataset maps transactional operational parameters via twenty key relational fields:

- Order Details: Order ID, Order Date
- Customer Details: Customer Name, Gender, Age
- Geographic Details: City, State
- Product Structure: Product Category, Product Name, Brand
- Financial & Order Metrics: Quantity, Unit Price, Discount Percent, Sales Amount, Cost Amount, Profit
- Fulfillment Channels & Sentiment ,Payment Method,, Sales Channel, Customer Rating, Returned.

3. Data Preprocessing Steps

The following systematic preprocessing and field configuration actions were implemented within Tableau before visualization construction began:

- Duplicate Elimination
- Data Type Assignment
- Geographic Role Configuration.

4. Sheets Created in the Report

4.1 Executive Overview Dashboard

Purpose: High-level view of overall business performance - Total Sales - Total Profit - Total Quantity - Average Customer Rating - Sales and Profit Trend by Month

4.2 Sales-Wise Performance Analysis

Purpose: Identify sales trends and distributions across dimensions - Sales Volume by Brand - Payment Method Distribution - Sales Amount vs. Discount % Scatter Plot - Product Level Sales Table

4.3 Profitability Analysis Dashboard

Purpose: Analyze profit distribution and identify cost drivers - Gross Margin % - Total Cost Base - Return Rate KPI - State Profitability Rank Chart - Category Profit Margins - Cost vs. Profit Breakdown

4.4 Key Insights & Recommendations Dashboard

Purpose: Provide data-driven strategic narratives and operational problem areas - Quality Leak Chart (Rating vs. Returns) - The Discount Dilemma Line - Logistics Risk Breakdown - Channel Capture Comparison

5. DAX Measures Used

- Gross Margin %: $\text{SUM}([\text{Profit}]) / \text{SUM}([\text{Sales Amount}])$
- Return Rate %: $\text{COUNT}(\text{IF } [\text{Returned}] = \text{'Yes'} \text{ THEN } [\text{Order ID}] \text{ END}) / \text{COUNT}([\text{Order ID}])$
- Total Sales : $\text{SUM}([\text{Sales Amount}])$
- Total Profit : $\text{SUM}([\text{Profit}])$
- Average Order Value (AOV): $\text{SUM}([\text{Sales Amount}]) / \text{COUNTD}([\text{Order ID}])$
- Discount Amount: $[\text{Unit Price}] * [\text{Quantity}] * [\text{Discount Percent}] / 100$
- Total Orders Processed: $\text{COUNTD}([\text{Order ID}])$

6. Key Insights & Findings

- Laptop & Accessories Performance: Laptops generate the highest overall sales volume and net profit (\$2,197,891.22), proving to be the primary engine of corporate growth.
- Lenovo Reverse Logistics Strain: Lenovo products exhibit a high return rate (53.3%), causing significant reverse-logistics overhead that impacts the brand's net margins.
- The Customer Satisfaction Deficit: The analysis reveals a clear pattern where 1-star product reviews strongly correlate with a high return rate (35.6%).
- Discount Margin Loss: Aggressive discounting practices reduce overall profitability, creating situations where product margins are heavily squeezed by high promotion rates.
- Distribution Network Value: Online channels deliver a higher average gross margin percentage (23.0%) than traditional brick-and-mortar storefronts (22.3%).

Assumptions

- Dataset records are accurate and complete.
 - Customer ratings correctly reflect satisfaction levels.
 - Return information is properly recorded.
 - Cost and profit values are calculated correctly.
 - External factors (inflation, competition, seasonality) are not considered.
 - Geographic and sales channel data are accurate.
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Recommendations

- **Implement Vendor Audits:** Launch an immediate quality and supply chain review of Lenovo product lines to address the high 53.3% return rate and cut down on reverse-logistics costs.
- **Establish Discount Caps:** Set a hard limit on flat-rate promotions at 15% to protect product margins from excessive discounting, shifting instead toward value-based product bundling.
- **Optimize Channel Strategy:** Focus marketing spend and inventory allocation on online channels to take advantage of their higher 23.0% gross margin capture rate.
- **Improve Product Descriptions:** Address the high return rate among poorly reviewed items by ensuring online product listings and marketing materials accurately reflect real-world specifications.

7. Final Conclusion (DA Report Summary)

- This Electronic Sales Data Analytics report provides an overview of sales performance, key profitability drivers, and return metrics across the business. By organizing clean transactional records into interactive, well-structured dashboard sheets, the organization can easily identify margin leakage points and isolate high-performing geographic markets. Moving forward, enforcing discount caps, improving vendor quality control, and shifting focus toward higher-margin online channels will help stabilize overall profits and improve operational efficiency.

End of Report